Exploratory Data Analysis of Used Cars

Understanding Price Patterns, Vehicle Characteristics, and Data Quality

Project Report

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- **Batch:** 2026, AI-MI(B)
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- **Course / Subject:** AI-ML (B)

- **Dataset:** cardekho-selected-columns.csv
- **Dataset Source:** https://www.kaggle.com/datasets/sukhmandeepsinghbrar/car-price-prediction-dataset

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1. Project Objective

The primary objective of this project is to perform a comprehensive Exploratory Data Analysis (EDA) on a used-car dataset (`cardekho-selected-columns.csv`) to uncover underlying patterns, evaluate data quality, assess missingness and anomalies, and analyze how various vehicle attributes relate to market selling prices.

To structure the investigation, the following 8 core research questions were selected for detailed analysis:

1. Which car brands and fuel types are most common?
2. How are used-car selling prices distributed?
3. How does vehicle age relate to selling price?
4. How does distance driven relate to selling price, and how is fuel efficiency associated with price?
5. Do automatic and manual cars differ in selling price?
6. Which numerical features are strongly correlated?
7. Which brands have the highest average selling prices?
8. Which records are unusually high or otherwise require further investigation?

2. Dataset Overview

The dataset consists of historical listings of used vehicles with attributes detailing physical characteristics, usage history, and selling price.

Feature Name	Description	Data Type
name	Car model/name	Object
year	Manufacturing year	Integer
selling_price	Selling price of the car	Integer
km_driven	Total distance driven	Integer
fuel	Fuel category	Object
seller_type	Type of seller	Object
transmission	Manual or automatic	Object
owner	Previous ownership category	Object
mileage(km/ltr/kg)	Fuel efficiency	Float
engine	Engine capacity	Float

Note on Sample Records: Initial inspection reveals diverse vehicle types ranging from budget hatchbacks to premium sedans, spanning various manufacturing years and fuel categories.

3. Analytical Approach

The analysis follows a systematic data science pipeline structured into four main phases:

1. **Data Quality Assessment & Cleaning:** Identifying missing values, duplicate entries, structural data type misalignments, and potential measurement units.
2. **Descriptive Statistical Analysis:** Computing measures of central tendency (mean, median) and dispersion (range, IQR) across key numerical and categorical features.
3. **Visual Exploratory Data Analysis (EDA):** Generating univariate distributions, bivariate relationships, and multivariate correlation structures to visualize interactions.
4. **Outlier & Anomaly Investigation:** Applying statistical bounds (IQR method) to isolate extreme listings and evaluate their domain legitimacy.

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4. Data Quality Assessment

4.1 Missing Values

A systematically structured audit of missing values was conducted across all attributes in the dataset.

Column Name	Missing Count	Percentage (%)
name	0	0.00%
year	0	0.00%
selling_price	0	0.00%
km_driven	0	0.00%
fuel	0	0.00%
seller_type	0	0.00%
transmission	0	0.00%
owner	0	0.00%
mileage(km/ltr/kg)	221	~2.7%
engine	221	~2.7%

- **Cleaning Decision:** Missing values in technical specification columns (`mileage` and `engine`) account for a minor fraction (~2.7%) of the total dataset. Depending on the visual context, these rows were handled via listwise deletion for complete-case visualizations or kept intact for non-engine univariate analyses.
- **Limitation:** Dropping these missing entries reduces the sample size slightly, but avoids introducing imputation bias into performance metric distributions.

4.2 Duplicate Records

An evaluation of complete row duplicates was performed across all fields.

- **Cleaning Decision:** Duplicate listings (exact matching rows across all attributes) were identified and dropped to prevent over-representation of repeated dealer listings or re-posted advertisements.

- **Limitation:** Some identical rows may represent genuinely distinct vehicles with identical specs and mileage; removing them assumes duplicate rows stem from system re-indexing or duplicate postings.

4.3 Outlier Detection and Treatment

Outliers were identified using the standard Interquartile Range (IQR) method:

$$IQR = Q_3 - Q_1$$

The lower and upper bounds are defined as:

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR$$

- **Boundaries:** Applied across `selling_price`, `km_driven`, and `engine` capacity. Extreme values above the upper threshold (e.g., high-end luxury vehicles) were flagged.
- **Important Limitation:** Removing statistical outliers blindly would strip away legitimate high-end luxury vehicles (e.g., BMW, Mercedes-Benz) from the dataset. Therefore, extreme values were analyzed contextually rather than automatically truncated.

4.4 Data Types and Categories

Data types were validated upon loading. Numeric variables (`year`, `selling_price`, `km_driven`) were stored as integers/floats, while categorical variables (`fuel`, `seller_type`, `transmission`, `owner`) were encoded as categorical string variables.

(Suggested Screenshot Note: Insert `df.info()` and `df.head()` output snippet here)

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5. Numerical Summary

Descriptive summary statistics were calculated to understand central tendency, spread, and range for continuous numerical variables.

Variable	Mean	Median	Minimum	Maximum
Manufacturing year	2013.8	2015.0	1983.0	2020.0

Variable	Mean	Median	Minimum	Maximum
Selling price (₹)	638,271.8	450,000.0	29,999.0	10,000,000.0
Kilometres driven	69,801.1	60,000.0	1.0	2,360,000.0
Mileage (km/ltr/kg)	19.4	19.3	0.0	42.0
Engine capacity (cc)	1,458.2	1,248.0	624.0	3,604.0

- **Notes & Observations:**
  - **Selling Price:** The significant gap between the mean (₹638,271) and median (₹450,000) indicates a strong right-skewness driven by high-value luxury cars.
  - **Kilometres Driven:** The maximum value of 2,360,000 km represents an extreme outlier that requires verification against vehicle age.
  - **Mileage:** Includes values of 0.0, which represent missing or unrecorded efficiency figures rather than physical reality.

6. Categorical Summary

Frequency distributions were evaluated for all categorical descriptors.

Feature Name	Unique Categories	Most Frequent Class	Frequency (Mode Count)
Brand*	32	Maruti	High relative market share
Fuel	5	Diesel / Petrol	Predominant share
Transmission	2	Manual	Significant majority
Seller type	3	Individual	Majority of listings
Owner	5	First Owner	Majority of listings
Car model name	High cardinality	Maruti Swift / Alto variants	Top individual listings

- **Notes & Observations:**
  - Brand was extracted from the first token of the `name` column.
  - The market is dominated by first-owner, manual-transmission vehicles, with Diesel and Petrol making up over 90% of fuel choices.

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7. Visual Exploratory Data Analysis

Chart 1. Distribution of Selling Prices

- **Purpose:** Assess market price spread and degree of skewness.
- **Notebook Cell:** Section 4.1, Cell 3.
- **Observations:** Highly right-skewed distribution. The majority of vehicles trade below ₹1,000,000, with a long tail extending up to ₹10,000,000 for luxury imports.

Chart 2. Distribution of Manufacturing Years

- **Purpose:** Inspect fleet age composition within the sample.
- **Notebook Cell:** Section 4.1, Cell 5.
- **Observations:** Highly left-skewed distribution. Most listings represent vehicles manufactured between 2010 and 2019, reflecting a relatively modern inventory.

Chart 3. Manufacturing Year vs Selling Price

- **Purpose:** Visualize price depreciation trends over model years.
- **Notebook Cell:** Section 4.2, Cell 2.
- **Observations:** Strong non-linear positive association. Exponential price appreciation is observed for recent model years (2017–2020), while older vehicles convergence towards a low baseline price.

Chart 4. Selling Price by Owner Category

- **Purpose:** Evaluate the impact of ownership history on retained value.
- **Notebook Cell:** Section 4.2, Cell 6.
- **Observations:** First-owner vehicles command higher median prices and display wider variance. Selling prices progressively decrease through Second, Third, and Fourth & Above owner categories.

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8. Visual EDA: Multivariate Analysis and Outliers

Chart 5. Correlation Heatmap

- **Purpose:** Examine linear relationship strength across all quantitative fields.
- **Notebook Cell:** Section 4.3, Cell 1.
- **Observations:** Strong positive correlation between `engine` capacity and `selling_price`. Strong negative correlation between vehicle age and `selling_price`. Moderate negative correlation between `mileage` and `engine` capacity.

Chart 6. Car Age vs Selling Price

- **Purpose:** Analyze functional depreciation dynamics directly.
- **Notebook Cell:** Section 4.3, Cell 3.
- **Observations:** Clear inverse relationship; value drops rapidly within the first 5 years of operation before leveling off into a plateau.

Chart 7. Distance Driven vs Selling Price

- **Purpose:** Observe the influence of vehicle usage on market value.
- **Notebook Cell:** Section 4.3, Cell 5.
- **Observations:** General downward trend in selling price as cumulative `km_driven` increases, though variance remains high at lower mileage levels due to vehicle brand differences.

Chart 8. Selling Price by Transmission Type

- **Purpose:** Compare pricing distributions across transmission technologies.
- **Notebook Cell:** Section 4.3, Cell 7.
- **Observations:** Automatic transmission vehicles display a substantially higher median price and interquartile range compared to manual transmission vehicles.

Additional Analysis: Fuel Efficiency vs Selling Price

- **Observation:** Raw mileage displays a weak or slightly counter-intuitive negative correlation with price because luxury performance vehicles often have lower fuel efficiency but significantly higher market prices.

- **Important Correction Note:** Zero-mileage values (0.0 km/ltr) reflect unrecorded technical data and should be masked prior to calculating efficiency trends.

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9. Key Findings, Limitations, and Conclusion

9.1 Key Findings

1. **Brand Dominance:** Maruti and Hyundai dominate market volume, together comprising the vast majority of economy listings.
2. **Fuel Preference:** Diesel and Petrol power over 90% of listed vehicles; CNG/LPG options form a niche segment.
3. **Price Distribution:** Prices are heavily right-skewed with a median of ₹450,000 and extreme premium outliers exceeding ₹5,000,000.
4. **Depreciation Rates:** Vehicle age is the single strongest negative driver of price, experiencing its sharpest decline within the first 5 years.
5. **Usage Impact:** Higher mileage (`km_driven`) correlates with lower price, though brand equity often overrides usage penalties.
6. **Transmission Premium:** Automatic vehicles hold a significant price premium over manual equivalents across comparable age brackets.
7. **Brand Tiering:** Premium brands (e.g., BMW, Mercedes-Benz, Audi) occupy distinct price bands far above volume market brands regardless of age.
8. **Data Anomalies:** A subset of listings contain extreme mileage figures (>1,000,000 km) or zero-efficiency ratings (0.0 mileage) requiring filtering.

Note on Brand Price Comparisons: Raw brand averages are heavily skewed by model mix; luxury brands average substantially higher baseline prices than mass-market producers.

9.2 Limitations

- **Missing Technical Data:** Approximately 2.7% of entries lack mileage and engine metrics.
- **Recorded Zero Values:** Several entries report 0.0 for fuel efficiency, skewing baseline calculations.
- **Lack of Regional Context:** Geographical location (city/state) is absent, ignoring regional resale price variations.
- **Condition Metrics Unrecorded:** Physical vehicle condition, accident history, and service records are not captured.

- **Duplicate Listings:** Presence of identical entries from bulk dealer re-posts required deduplication assumptions.
- **Right-Skewed Outliers:** IQR truncation risks discarding genuine luxury car market dynamics.
- **Static Snapshot:** Data represents a static temporal snapshot without longitudinal price tracking across time.

9.3 Conclusion

This Exploratory Data Analysis successfully parsed, cleaned, and evaluated the `cardekho-selected-columns.csv` dataset. The analysis highlights that used-car prices in this market are predominantly governed by vehicle age, engine capacity, transmission type, and brand tier. While the data exhibits good overall integrity, proper handling of missing specs, zero-mileage entries, and high-value luxury outliers is essential before deploying predictive machine learning models.